Attune for You: Exploring Solarpunk Through Decolonial Design

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Submission Id: 353

ABSTRACT

Attune for You, is an ongoing role-playing game project that explores solarpunk beyond its visual style. Although solarpunk is often recognized through bright natural environments and sustainable technologies, this project focuses more directly on its social and political ideas, especially decolonial thinking, consent, repair, autonomy, and coexistence. Set on Mars, the game follows Escropita, a human volunteer who works with the martians called the Mycaris through fungal auditory communication rather than colonial control or technological domination. The game combines a 3D world with 2D character assets and includes systems centered on limited energy use, infrastructure repair, and player agency and NPC's consent, and audio generated from living mushrooms. This paper describes the game's design, its connection to solarpunk, and the ways its mechanics, visuals, and audio aim to express the genre's broader values beyond its aesthetic norms. Finally, we outline a planned user study to examine how players interpret these design choices and perceive solarpunk themes through gameplay.

CCS CONCEPTS

• **Applied computing → Computer games**.

KEYWORDS

Indie, Game Genre, Serious Games, Solarpunk, Eudaimonia Games, Green Games, Game Design, Iterative Game Design, De-colonial Game Design, Ethical Games

ACM Reference Format:
Anonymous Author(s). 2026. Attune for You: Exploring Solarpunk Through Decolonial Design. In . ACM, New York, NY, USA, 4 pages. https://doi.org/XXXXXXX.XXXXXXX

1 INTRODUCTION

In response to the global climate crisis, designers and scholars are using games to educate players about ecological concerns as part of a trend referred to as environmental gaming [5, 15]. Simultaneously, there is growing interest in speculative climate fiction genres like solarpunk in the HCI research community [4, 19]. Solarpunk is recognizable through optimistic depictions of the future, which feature technology that supports social interests without compromising the natural world. Since the early 2000s, solarpunk has been explored in literary and audiovisual media [13]. More recently, games are exploring themes related to the genre, such as *I Was a Teenage Exocolonist* [18], *Terra Nil* [7], and the upcoming *SOLARPUNK* [3]. Such games borrow solarpunk aesthetics by depicting near-futuristic scenarios in brightly colored natural environments embedded with technology. That stated, many of the video games that make up the emerging solarpunk canon are faithful to the genre's audiovisual conventions, but do not fully reflect the genre's accompanying philosophy, which imagines futures defined by care-based ethics, diversity, and decolonial, post-capitalistic systems [20].

Thus, our design team is attempting to center solarpunk's philosophical orientations in a game. To do this, we are subverting the genre's typical representations as a way to decenter the importance of aesthetic fidelity. Instead of depicting green, earthy environments interspersed with sustainable technology, our game takes place on a red-hued, abandoned colony on Mars. The audiovisual design is intended to be psychedelic, which goes against the genre's pleasing visual iconography. This design choice is informed by cautions found in utopian scholarship, which notes that the inoffensive, tranquil solarpunk aesthetic is vulnerable to appropriation by corporate interests attempting to "greenwash" extractive agendas [8]. The technology presented in the game is depicted as occasionally inconvenient and unsuitable for solving major problems. The decision to present technology this way is informed by criticism of solarpunk titles that foreground advanced technology, and not communal engagement, as the central approach to solving ecological problems [14]. Finally, the player-character in our game must attempt to communicate and understand others, which connects to the philosophy's emphasis on collective action, as explained in *A Solarpunk Manifesto*, a popular document frequently referred to in solarpunk forums [17].

Once the game is completed, we intend to conduct a user study to assess the legibility of our subversive design choices, which were informed by Flanagan's critical play method [6]. One challenge of subverting an emerging, relatively minor genre is that players may not be familiar enough with the generic conventions to accurately recognize how and why they are being critiqued. Additionally, there is a risk that subversion could unintentionally undermine solarpunk's messaging; this outcome does not align with our supportive stance on solarpunk as a generative artistic and political concept. For these reasons, a future study will examine whether players are able to perceive an inspiring vision of solarpunk regardless of prior familiarity, and whether our subversive design choices made the genre's ideas visible. Ideally, the study will help us identify how subversive strategies can be used as tools for emphasizing foundational concepts rather than merely disrupting expectations, particularly for designers working with burgeoning, politically oriented game genres.

2 ATTUNE FOR YOU

This section introduces our ongoing game project, *Attune for You*, and discusses how its design aligns with the solarpunk genre.

Conference'17, July 2017, Washington, DC, USA

ACM ISBN 978-x-xxxx-xxxx-x/YY/MM
https://doi.org/XXXXXXX.XXXXXXX

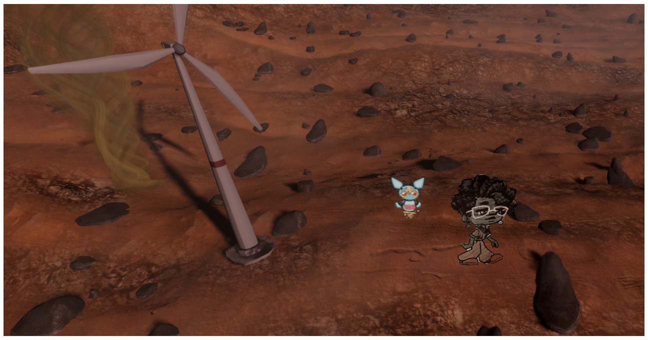

Figure 1: Screenshot from under development build of *Attune for You*. The image shows Escropita and Daffodil as a 2D character placed within the game's 3D Martian world.

2.1 Game Overview

Attune for You is being developed in Unity 6.2 and features a 3D world with 2D character assets. The game begins with a prologue introducing the main character, Escropita, and her mission on Mars. Upon arrival, she encounters the abandoned remnants of Joy-Tech facilities and meets the Mycari, native Martians who live underground and do not rely on Earth-based life support systems to survive.

To communicate with the Mycari, Escropita must engage with Mars's fungal network. She can seek the Mycari's help only after guiding them to a musical mushroom grove. There, the Mycari create a musical pattern that the player must mimic to establish successful communication. Once the interaction succeeds, a Mycari will follow the player, although its destination depends on that individual Mycari's personality.

The Mycari have three personality-based agent behaviors: some assist with repairs, some deconstruct structures, and others wander aimlessly. Although they are visually identical, Escropita's AI companion, Daffodil, can identify their traits through scanning. Daffodil can also scan Joy-Tech infrastructure to assess its condition. However, scanning consumes a significant amount of energy, so the player must use Daffodil strategically. When Daffodil runs out of power, it becomes unresponsive until it is fully recharged through solar energy.

The Joy-Tech life support system depends on electricity generated by wind turbines. Randomly timed tornadoes can damage these turbines and disrupt energy production. Escropita can work with the Mycari to repair the windmills and keep critical systems running. Some obelisks, built by the Joy-Tech exo-colonialists, were originally used to monitor and surveil the Mycari. Because surveillance is not essential, the player can choose to help certain Mycari dismantle these obelisks and redirect power to the life support system. If the life support system fails, the mission is lost.

Our role-playing game is set in a future where humanity has overthrown the billionaire class and entered a new era defined by social equity, sustainability, and peace. The player takes on the role of Escropita, a member of a volunteer group that travels across the galaxy in repurposed spacecraft to offer reparations on behalf of humanity. She arrives on Mars in response to a request for help from the Mycari, an alien species living there. On the planet, she finds the decaying remains of Joy-Tech, a human megacorporation that colonized nearby worlds centuries ago. The Mycari, who do not use verbally expressed language, encourage Escropita to communicate through musical rituals involving native fungi. Alongside her sustainably powered AI companion, Daffodil, Escropita must learn to understand what the Mycari want by attuning herself to the mushroom music they create together.

The game's 2D-on-3D visual design is shown in Fig. 1, where Escropita appears as a 2D character within the 3D Martian environment.

2.2 Solarpunk Design

Attune for You draws some of its core ideas of solarpunk from *A Solarpunk Manifesto* and the Solarpunk Wiki [1, 17]. Here, solarpunk is not just a visual style. It is also about care, repair, community, ecological balance, and living together in cooperation. For that reason, the game does not present a polished green future built only on advanced technology. Instead, it focuses on consent, coexistence, limits, and the use of systems already present in the world. These ideas appear in how Escropita communicates with the Mycari, how Joy-Tech's remains are reused or dismantled, and how technology is treated as something that must work with biological and collective life rather than control it.

Smart Citizenry. Escropita does not solve problems through code, wires, or software. Instead, she communicates through the fungal network and works through forms of connection that are already biologically present in the world. In this sense, the game rejects the logic of the "smart city" and instead imagines a form of smart citizenry grounded in local knowledge, ecological attunement, and interdependence.

Anti-Hierarchical Relations. Escropita is not a leader, ruler, or savior. She enters an existing Mycari civic system as a participant and must learn to coexist with the Mycari as equals. This means learning how they communicate through the fungal network and adapting to them, rather than forcing them to respond on human terms.

Post-Capitalism. Joy-Tech is not framed as a symbol of progress. It represents the failed legacy of billionaire-led space colonization: abandoned infrastructure, broken contact, and ecological harm. The game takes place after that failure, in a world still living with its consequences.

Decolonialism and Consent. Through subversion, decolonial game design can offer social critique through virtual worlds [21]. By positioning players within subaltern perspectives, such work may challenge the Eurocentric assumptions that continue to shape mainstream media [2]. Similarly, recent research on consent in game studies argues that consent practices are often weaponized by those against "wokeism," while feminist and queer perspectives instead frame consent as an ethical orientation rooted in relational interaction in games [16]. These closely related interventionist perspectives highlight how political and ethical ideas in games are often reduced to surface-level interpretations and suggest the need for design approaches that deeply reckon with these ideas.

(a) Escropita (b) Daffodil (c) Mycari

Figure 2: Character design – *Attune for You.*

Solarpunk is closely tied to decolonial thinking, and the game reflects this by refusing the familiar narrative of a human outsider arriving to fix a broken world. Instead, it centers the autonomy of the Mycari, their existing systems, and their own ways of relating to one another and to their environment. Consent matters here: the Mycari may choose not to help, and Escropita must accept that. Repair does not come from domination, but from respecting limits, refusal, and difference.

Repair Culture and Repurposing. The game centers repair over replacement. Joy-Tech is abandoned infrastructure from a failed human project, while Escropita belongs to a volunteer group that travels in repurposed spacecraft and works with what already exists. The Mycari must decide what to do with Joy-Tech's remains: preserve them, adapt them, or dismantle them.

Appropriate Technology. The game does not reject technology outright, but it treats technology as something that must operate within ecological limits. Daffodil runs on solar power and becomes unusable when that energy is unavailable. Likewise, the communication system avoids standard human interfaces such as screens or keyboards. Instead, Escropita must work through the fungal network and engage with the Mycari on their own terms. Technology and problem-solving are therefore framed as practices that work with the environment rather than above it.

Utopian Orientation. The game remains aligned with solarpunk's utopian orientation. It imagines a future after crisis rather than a future defined entirely by collapse. Earth did not end in ruin; it changed course. The world of the game is shaped by repair, shared futures, and new forms of communication that challenge hierarchical rule.

Visual Design. The game's solarpunk aesthetic is expressed through its 2D characters placed within a 3D world (Fig. 2). Escropita (Fig. 2a), Daffodil (Fig. 2b), and the Mycari (Fig. 2c) do not appear sleek, metallic, or hyper-industrial. They look drawn, textured, and tied to living systems. Daffodil makes technology feel small and companion-like rather than dominant, while the Mycari visually connect the world to fungi, soil, and local ecology. The texture style, inspired by *Chowder*, uses moving patterns within the characters, making them feel handmade rather than mass-produced.

Audio Design. Fungi have been explored in HCI scholarship in a variety of ways in recent years. In speculative design research, mycelial networks are a useful metaphor for considering potential collaborative, distributed systems [10]. In tangible interaction spaces, fungi have been used as a sustainable biomaterial for design work [9]. Some research also explores fungi as a kind of collaborator, and not just a usable resource, through the lens of biophilic entanglement [4]. This scholarship showcases the utility of fungi in design thinking, and our game design process builds upon this conceptual work by incorporating mushrooms into the design. Mushrooms are not a metaphor or material, but an important entity in the virtual world; the Mycari create auditory cues with mushrooms as a communication ritual, and the player must learn how to interact with them in the same way, which represents ecological interdependence.

Figure 3: Using probes to record biological data from mushrooms, which is then converted into MIDI tracks on the MPC.

The game's audio also comes from living fungal systems. We record data from real mushrooms with non-invasive probes, use it as sheet music, and input it as MIDI tracks through an MPC in order to create musical tunes with the recorded sounds. Figure 3 shows a wild mushroom being probed for signal. This grounds the soundtrack in the biological world itself rather than relying only on synthetic effects. As a result, the audio remains tied to the same living systems that shape the game's world and its approach to communication.

3 PLANNED USER STUDY

Once the game is complete, we plan to conduct a user study to assess whether players can recognize the game as solarpunk despite its subversive characteristics. We also aim to explore how players interpret the game's ideas related to consent, ecological harmony, and decolonialism. Participants will be students at Worcester Polytechnic Institute, between the ages of 18 and 24, who have varying levels of familiarity with solarpunk and experimental indie games. Since the study focuses on interpretive understanding rather than performance or enjoyment, the questions will primarily be open-ended. Participants will not be explicitly informed about solarpunk, decolonialism, or consent before gameplay. This approach ensures that their responses reflect the legibility of these ideas through the game's design itself, rather than their ability to apply terminology to their impressions. An optional debrief may follow to explore whether participants observed these themes once they are introduced.

Additionally, we are designing the game to be responsive to player agency. Joshi et al. found that giving players more agency increased enjoyment and intrinsic motivation without reducing learning outcomes [11, 12]. Players receive different endings depending on how they respond to the nonverbal Mycari; the most cooperative behavior is rewarded with mutual information exchange, and the most human-centered behavior leads to less information exchange and negative interstellar relations between Earth and Mars. Therefore, we will ask participants to reflect on the outcomes of their behavior in-game to determine if the system's responses inform their understanding of consent, ecological harmony, and decolonial relations.

To capture how players interpret the game's design values and generic qualities, we propose the following open-ended questions:

- What role do humans seem to play in the world of the game?
- Did the game feel optimistic, critical, or something in between? What made it feel that way?
- What values do you think the game is trying to communicate?
- Did the game reward the kinds of actions you expected? Why or why not?
- How would you describe the relationship between the player and the Mycari?
- What genre or type of game did this feel like to you?

Responses to these questions will help clarify two related inquiries: whether subversions of minor genres in games confuse players or enhance understanding, and whether decolonial, consent-forward design strategies effectively communicate non-human autonomy. This study is intended to provide a method for design researchers to evaluate speculative genres and to offer insight into how players interpret philosophical ideas through decolonial mechanics.

4 CONCLUSION

Solarpunk is a genre that imagines hopeful futures shaped by ecology, care, and collective life [19]. Previous work focus mainly on its visual style, such as green environments, bright colors, and sustainable technology [13]. However, the social and political side of solarpunk has been less emphasized. This includes ideas such as decolonial thinking, consent, repair, re-purposing, coexistence, and ways of living that move beyond extraction and domination [8, 20].

Attune for You builds on that broader side of solarpunk. The game follows Escropita on Mars, where she must work with the Mycari through fungal communication. Instead of solving problems through control or technology, the game focuses on autonomy, repair, consent, and living together on shared terms. In this way, the project treats solarpunk as more than a visual style. It uses the genre to explore social and political ideas, while also deviating from its aesthetic language. Once Attune for You is complete, a user study will be conducted to examine how players perceive and interpret these subversive elements, whether they recognize the game as solarpunk, and how their interactions with the systems influence understanding of its philosophical, decolonial themes. This approach allows us to evaluate how our experimental design strategies communicate genre and philosophical values.

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